
Q1 — Dimensional Analysis (Model Discovery)

Solution 1: Dimensional Analysis

Step 1: Assign Dimensions

The physical dimensions for the variables are:

- $[T] = T^1$ (Time)
- $[m] = M^1$ (Mass)
- $[L] = L^1$ (Length)
- $[g] = L^1 T^{-2}$ (Acceleration)

Step 2: Set up the Dimensional Equation

Using the formula $T = k m^a L^b g^c$ where k is a dimensionless constant:

$$M^0 L^0 T^1 = (M)^a (L)^b (L T^{-2})^c$$

Combining terms:

$$M^0 L^0 T^1 = M^a L^{b+c} T^{-2c}$$

Step 3: Solve for Exponents by Comparison

- **Mass (M):** $0 = a \implies \mathbf{a = 0}$
- **Time (T):** $1 = -2c \implies \mathbf{c = -1/2}$
- **Length (L):** $0 = b + c \implies 0 = b - 1/2 \implies \mathbf{b = 1/2}$

Resulting Model: $T = k m^0 L^{1/2} g^{-1/2} \implies \mathbf{T = k \sqrt{\frac{L}{g}}}$

Interpretation: In model discovery, this result proves that mass is an **irrelevant feature**. If an algorithm (like Symbolic Regression) were to analyze pendulum data, dimensional constraints prevent it from incorrectly assuming the period depends on mass.

Q2 — Kinematics (Trajectory Model)

Solution 2: Particle Kinematics

(a) Velocity and Acceleration Functions

By differentiating the position function $x(t) = 4t^3 - 3t^2 + 2t$:

- $v(t) = \frac{dx}{dt} = 12t^2 - 6t + 2$
- $a(t) = \frac{dv}{dt} = 24t - 6$

(b) Real Roots for Velocity

Setting $v(t) = 0 \implies 12t^2 - 6t + 2 = 0$.

Using the discriminant $D = b^2 - 4ac$:

$$D = (-6)^2 - 4(12)(2) = 36 - 96 = -60$$

Since $D < 0$, there are no real roots. The velocity **never becomes zero**.

(c) Motion at $t = 1$ s

$v(1) = 12(1)^2 - 6(1) + 2 = 8$ m/s (Positive direction)

$a(1) = 24(1) - 6 = 18$ m/s² (Positive direction)

Justification: Since velocity and acceleration share the same sign, the acceleration is acting to increase the magnitude of velocity. Thus, the particle is **speeding up**.

Solution 2: More explanation

A positive acceleration does not automatically mean the body is speeding up, it depends on the direction of velocity relative to acceleration.

Key points:

- Acceleration is the rate of change of velocity.
- If velocity and acceleration have the same sign (both positive or both negative), the speed increases $\rightarrow$ the body is speeding up.
- If velocity and acceleration have opposite signs, the speed decreases $\rightarrow$ the body is slowing down.

Velocity $v(t)$	Acceleration $a(t)$	Result
$v(t) > 0$	$a(t) > 0$	Speeding up in the positive direction
$v(t) < 0$	$a(t) < 0$	Speeding up in the negative direction
$v(t) > 0$	$a(t) < 0$	Slowing down (positive motion, negative acceleration)
$v(t) < 0$	$a(t) > 0$	Slowing down (negative motion, positive acceleration)

Solution 2: More explanation

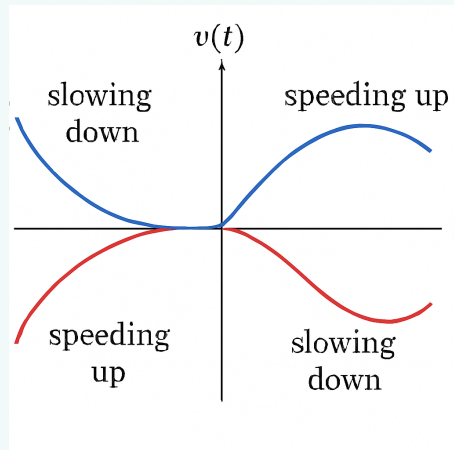


Figure 1: Velocity–Acceleration Quadrants in Rectilinear Motion

- **Quadrant I:** $v > 0, a > 0 \Rightarrow$ Speeding up forward.
- **Quadrant II:** $v > 0, a < 0 \Rightarrow$ Slowing down forward.
- **Quadrant III:** $v < 0, a < 0 \Rightarrow$ Speeding up backward (magnitude of velocity increases).
- **Quadrant IV:** $v < 0, a > 0 \Rightarrow$ Slowing down backward.

This is why we often say: acceleration tells you how velocity changes, but velocity itself tells you the direction of motion. Both together determine whether the object is gaining or losing speed.

Q4 — Projectile Motion & Parameter Estimation

Solution 4: Parameter Estimation

(a) Time of Flight (T):

Set $y(t) = 0 \implies v_0 \sin \theta t - \frac{1}{2}gt^2 = 0$. Solving for t (ignoring $t = 0$):

$$\mathbf{T} = \frac{2\mathbf{v}_0 \sin \theta}{\mathbf{g}}$$

(b) Least-Squares Method:

To estimate v_0 and θ from observations (t_i, x_i, y_i) , we minimize the sum of squared errors (S):

$$S(v_0, \theta) = \sum_{i=1}^n \left[(x_i - v_0 \cos \theta t_i)^2 + (y_i - (v_0 \sin \theta t_i - \frac{1}{2}gt_i^2))^2 \right]$$

(c) Linearity and Impact:

The parameters v_0 and θ are **nonlinear**. Unlike linear regression, this requires numerical iterative solvers (e.g., Levenberg-Marquardt). Nonlinear estimation is more sensitive to "initial guesses" and local minima.

Q5 — Rotating Wheel and Hanging Mass

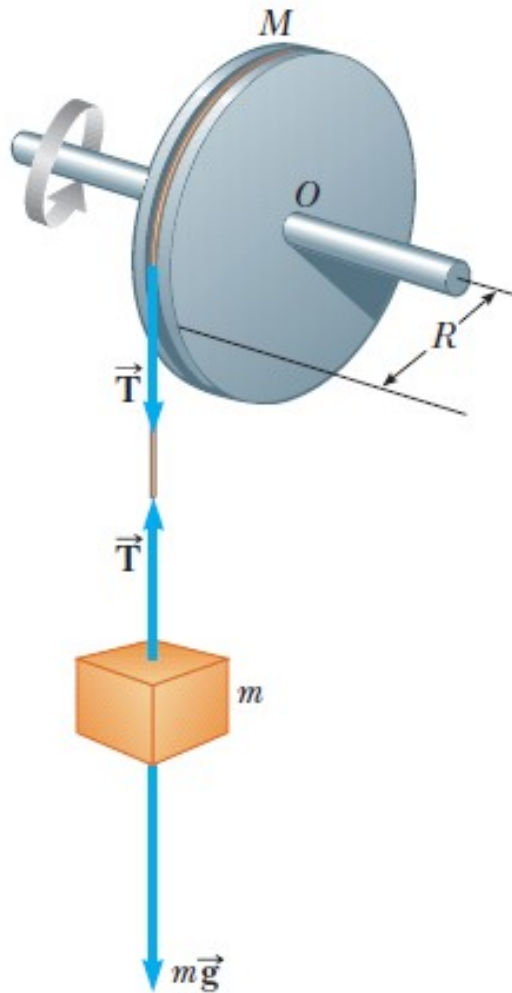


Figure 2: Sketch of the wheel system showing radius R , Tension T , and Mass m .

Solution 5: Rotational Dynamics

1. Equations of Motion:

- Mass (Translation): $mg - T = ma$
- Wheel (Rotation): $\tau = TR = I\alpha$
- Constraint: $a = R\alpha \implies \alpha = a/R$

2. Solving for Acceleration (a):

Substitute $T = \frac{Ia}{R^2}$ into the mass equation:

$$mg - \frac{Ia}{R^2} = ma \implies mg = a \left(m + \frac{I}{R^2} \right)$$

$$a = \frac{mgR^2}{mR^2 + I}$$

3. Final Results:

- Angular Accel: $\alpha = \frac{mgR}{mR^2 + I}$
- Tension: $T = \frac{Img}{mR^2 + I}$